

Pulse Memorial: A web broadcasted multichannel sound installation exploring queer grief through audience participation

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Abstract

The "Pulse Memorial" is a participatory art installation exploring multichannel web audio streaming, data sonification, and field recordings to address queer grief and erasure. Inspired by Toni Morrison's concept of "disremembrance," it offers an immersive sonic experience fostering collective remembrance. Sound is generated by combining data sonification and field recordings into an 8-channel score, then transmitted via WebRTC to participants—one channel per device—for a multi-channel sonic event. This project transforms memorials from passive, site-specific events to community-driven experiences. It provokes curiosity about digital media's role in preserving cultural memory and amplifying marginalized voices.

CCS Concepts

• Networks; • Information systems; • Hardware;

Keywords

queer memorial, field recording, multichannel, network music

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1 Introduction

The Pulse Memorial is a web-streamed sound installation that commemorates the lives lost in the mass shooting at the Pulse Bar in Orlando, Florida, where a gunman killed forty-nine Latine, Black, and queer patrons. This tragedy happened in June 2016 during a period of heightened fear among immigrant and marginalized communities anticipating the Trump election victory in November. Inspired by Toni Morrison's concept of "disremembrance," which denotes an emotional experience of erasure due to a lack of acknowledgment, this web broadcast disrupts disremembrance by using sound to evoke presence. The installation commemorates victims that continue to be erased out of broader histories of domestic terrorism and gun violence.



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Participants can load the Pulse Memorial project on their mobile devices at <https://pulse.memorial/>. Upon entering the site, participants will receive one of eight continuously broadcasted tracks at random. The sound that the server draws from is generated by a combination of field recordings and audio created through data sonification (more details provided in the Implementation section). We used the number of victims and their ages for the sonification process, as these were the only available data points. To that end, the work participates in spiritual traditions of the past which saw numbers and music as conduits for "the hidden correspondences of nature." The term "speculative music" is used to denote these more esoteric models and theories of harmonies [Godwin, 1995].

Together as one listening unit, participants move around each other to form a moving multi-channel playback of sound-in-space. This project introduces a dual framework that views the web as both a broadcasting medium and a social network connecting communities. The web-based broadcast system utilizes global networking for local interactivity, transforming listening into an active and engaging experience. While one could "visit" the memorial alone or through playback of the website on multiple devices, visiting as a group makes the most sense for our intended purpose of communal remembrance.

Mobile devices serve as a crucial starting point for creating virtual, local, and unique spaces of connection as every mobile device might have particular defects, wear and tear, or sound enhancements that unsettle the fallibility of permanence in sculptural, physical memorials. The Pulse nightclub, a physical space still standing, also functions as a place for ad-hoc community and connection, as it currently is plastered with DIY banners, low resolution photographs, trinkets, and non-permanent ephemera that speaks to larger political marginalization in Florida, a state well-known for its public and insidious attack on queer policies. The installation's web-based format mirrors the nightclub's sense of collective repair by networking both locally and at the server level. People often turn to their phones for comfort in unfamiliar or uncomfortable environments. This installation leverages that tendency, creating local networks where participants stream various tracks from the server, each contributing unique sounds to the overall cacophony of streamed audio. During the initial demo, the dark space was illuminated by the light from the participants' phones, providing visibility for both individuals and the surrounding group, acting as digital candlelight, popularly seen at outdoor, public vigils.

2 Related Works

We situate our work alongside and in relation to several influential figures, drawing inspiration from Fluxus, Network Music, Actor-Network Theory (ANT), histories of hermeticism, and experimental



Figure 1: Pulse Memorial and participants in state of listening and reflection

art. Critical contributors like Phil Kline [Gann, 1995], Anna Xambó [Xambó et al., 2019], Gerald Roma [Xambó and Roma, 2020], Golan Levin [Levin, 2005], Benjamin Taylor [Taylor, 2017] and Jason Freeman [Freeman, 2005] have shaped our understanding of network music, participatory happenings, data sonification, and memorials. We aim to inform and elevate the Pulse Memorial project by exploring these artists' contributions.

The Pulse Memorial project builds upon the history of network music, which emphasizes collaborative compositions facilitated by mobile technology. A significant component of network music that our project incorporates is the participatory approach, utilizing the audience's mobile devices to stream audio and transform them into a unified, one-of-a-kind instrument or spontaneous multichannel ensemble. In this respect, we share certain affinities with the eccentricities of Luigi Russolo, an iconoclastic member of the Italian futurist movement whose self-constructed noise instruments (the *intonarumori*) was pivotal in our understanding of self-made musical objects that enhance a viewer's cognitive abilities [Chessa, 2012]. This method contrasts with traditional composer-centric models, reflecting Xambó's influence in promoting shared collective listening and empowering participants with control over sound playback [Xambó and Roma, 2020]; [Oliveros, 2005]. This active engagement renders the audience an active participant, even a composing leader, streaming commemoration about the victims via sonic translation, abstraction, and chance operations.

Although our project has a digital modality, site specificity is significant, not to highlight one specific site but rather the everyday nature of coming together locally with a variety of handheld, portable devices. This everyday connection draws inspiration from Fluxus Happenings and Laurie Anderson's "Car Horn Symphony"

(1969). Like Anderson's orchestration of car horn synchronizations, our approach blends instructional guidance with spontaneous interaction to develop temporary, impactful experiences.

A key component of our project is performance, drawing on Diana Taylor's concept of slow performance which emphasizes practices unfolding gradually through repetition, ritual, and the slow accumulation of meaning [Taylor, 2017]. The Pulse Memorial piece embodies what Rebecca Schneider termed, "re-live," creating a generative experience in the present while leaving media traces for future iterations [Schneider, 2011]. As a web-based performance, it transcends physical and temporal boundaries, extending beyond heteronormative family archives into public spheres of shared memory, chosen family, and claims of "queer time" and "queer space" [Gratch and Gratch, 2021]; [Zucker and Simon, 2020]; [Halberstam, 2005]. Its accessibility on online platforms fosters collective memory, where trauma is shared, processed, and discussed, allowing survivors of the Pulse shooting and other U.S. American episodes of gun violence to connect and contribute to a broader cultural understanding of trauma [Menyhért, 2020].

3 Implementation

The project combines three main components: data sonification, a speculative soundscape of each individual audio channel, and multichannel web broadcast.

3.1 Sonification

We crafted sound through data sonification¹, using available data on victims' numbers and ages to compose original musical scores.

¹code for data sonification: <https://github.com/bvann9105/data-sonification->



Figure 2: Mobile browser and mobile browser view when user clicks play



Figure 3: An early demo revealed a solemn candlelight aesthetic

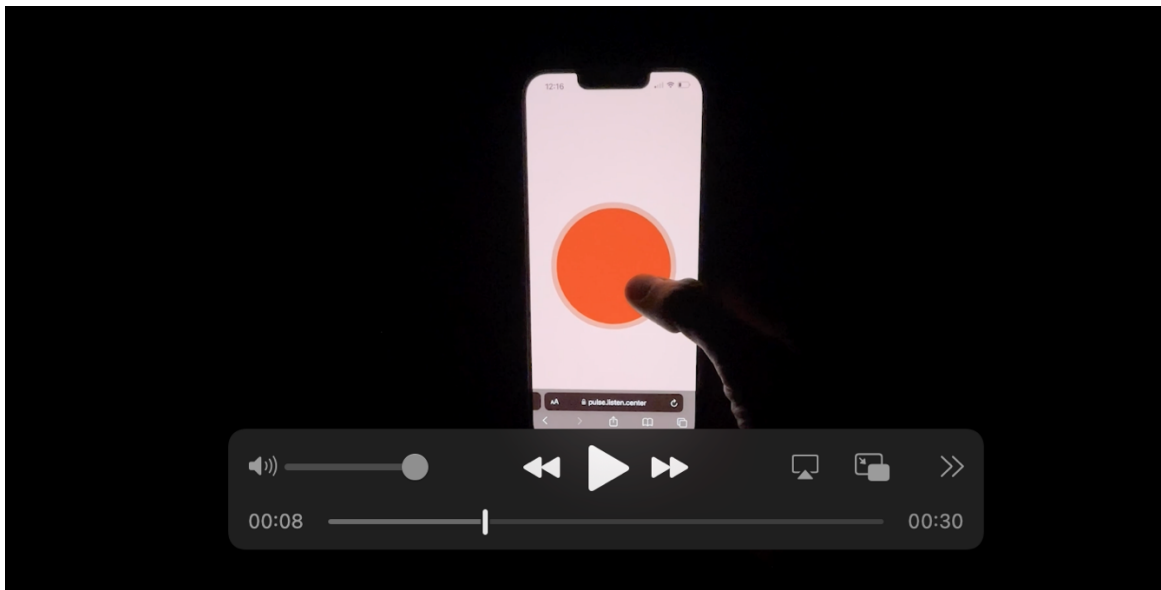


Figure 4: Binaural documentation of project (see supplemental material 1)

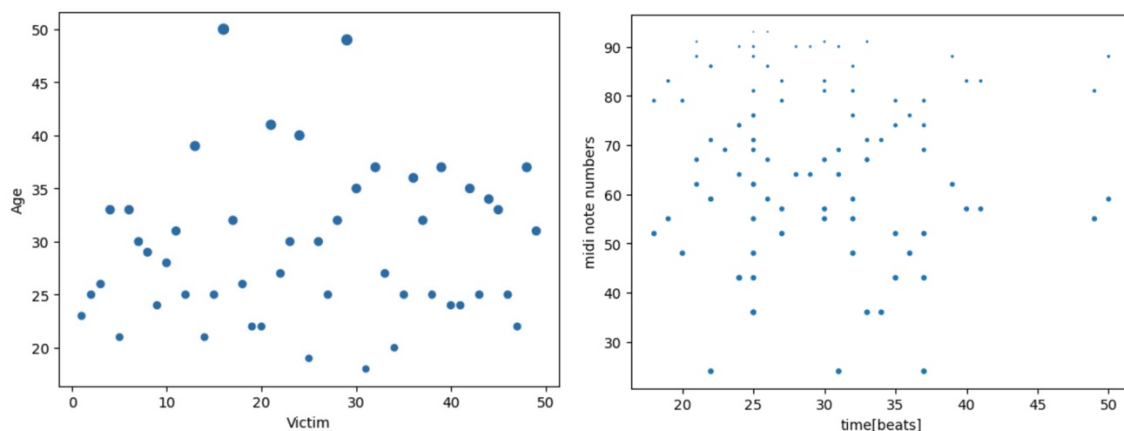


Figure 5: Two scatterplots of the Pulse data plotting victim by age and mapping data to midi

The data were scaled, translated into the MIDI scale, and arranged into a single soundscape.

Initially, the data for the Pulse Memorial were archived in a spreadsheet, lacking accessibility and emotional impact. Transforming it through data sonification made it more resonant and impactful. The process involved understanding the data, identifying gaps, observing trends, and interpreting their significance. Figure 4 shows the age distribution of victims, mostly between 20 and 37 years old. In designing the scale and parameters to translate the data into sound, we utilized several octaves of the pentatonic scale. We chose this scale for its ability to convey dissonance, effectively capturing the essence of loss and absence within the memorial's composition.

3.2 Soundscape

The Pulse Memorial uses a sculptural approach to arrange MIDI files, incorporating layering, collage techniques, and blending with field recordings. Our aim is not simply to adhere strictly to the data but to craft an imaginative soundscape that encapsulates the essence of grief, especially its everyday manifestations, which are often overlooked but deeply felt; near enough as the proximity of our hands to our phones.

Emphasizing the everyday aspect enabled us to integrate field recordings. Sound selection was guided by the impact, reflecting the aftermath of the mass shooting. The soundscape spans across the 8 tracks, creating an interplay with moments of synchronization and desynchronization. Each listener perceives nuances differently:

one might hear a faint, higher-pitched ringing, another an increasing tap of bell-like tones. Some tracks are processed with reverb to evoke a stronger sense of dissonance and loss. Field recordings include sounds like bus rides and coffee making, symbolizing the daily embodiment of grief. Electronic sounds in the composition represent grief and the presence of the 49 victims, while the field recordings capture everyday life. Combined, they evoke the experience of daily grief—sometimes, the electronic sounds and grief are prominent, while at other times, the sense of grief is subtler, and the field recordings take precedence.

The abstraction of the sound reflects the void surrounding what exactly transpired between 2:02 AM and 5:15 AM. This abstraction stands in for the gap left by the tragic event, in which 49 victims were killed—one of the deadliest mass shootings in U.S. history [Zambelich and Hurt, 2016]. The shift away from specificity also parallels the lack of concrete details or even basic information about the 49 victims.

Two participants in the initial demo characterized the sonic experience in the following way:

"I heard a variety of sounds that instilled different thoughts and textures into the auditory experience. Some sounds were reminiscent of dripping water and footsteps in an echoing cave, the white noise of crowded public transportation, and otherworldly metallic sounds reminiscent of a hand pan or similar instrument. The overall listening experience felt quite existential in nature. Hearing sound from every direction around me made me aware that the sound my own phone was creating was just one of many in the room. I relished walking around, taking in all the different sounds from peers and strangers I passed in the room."

"To me, the overall sound brings a chaotic sense of tension and release. Not being able to predict the next sound or where it will come from created an unpredictable atmosphere. The place where you're standing in the room could quickly shift from eerie calm to rhythmic unease, depending on who is around you and the sound coming from their devices. This mirrors daily interactions, where each person has their own unique sound at a given time. Sometimes these sounds are recognizable, while other times they are completely different. It's a strong reminder of the importance of listening to every sound in the room, not just your own."

3.3 Multichannel Web Broadcast

Participants load the project on their mobile devices in one room or setting, each receiving and playing one of eight possible WebRTC streams randomly assigned by the server. The collective acoustic experience forms a blended in-situ sonic environment with acoustic, transmissive, and situational characteristics.

The memorial produces a mobile playback² swarm characterized by its low-fi, untethered nature. Each participant's mobile device exhibits different auditory fidelity and acoustic responses, typically favoring mid and high-range content. The broadcast and participants remain unfixed, resulting in a dynamic field of sonic activity that shifts as participants move around. This multichannel quality contrasts with traditional static listening experiences, as sound emanates from various directions, creating a sense of multichannel movement with mobile speakers. Furthermore, as tracks overlap in

space, we introduce sub-second latencies via WebRTC and participants' playback devices, creating a choral effect that enriches the experience with depth and resonance.

Moreover, the memorial leverages global networking to foster hyper-local togetherness, emphasizing its transmission and situational components to create an immersive experience. Experiencing it in person and aloud with others is best, evoking an intimate and somewhat awkward atmosphere akin to theater but with an implicit acknowledgment of connecting through sound. This listening experience takes on a performative quality, requiring active engagement as participants become communal inhabitants of the memorial.

4 Discussion and Future work

In developing this iteration for the web's immaterial space, we made several noteworthy considerations regarding technology, format, and implementation. One consideration for this web-based project is that the web serves as a place where queer identities can flourish.

The internet provides queer individuals with unprecedented opportunities for community-building, self-expression, experimentation, and self-publishing, enabling them to shape their queer identities locally and globally [Williams et al., 2015]. Platforms like social media allow individuals to explore deviant behaviors in society, combating social isolation and fostering connections [Ferreday and Lock, 2007]. The project's sonic reimagining not only brings forgotten subjects from the past into focus but also highlights those slipping away from memory today [Richardson, 2013]. By introducing new elements into existing narratives surrounding the Pulse shooting, reimagining enables us to transform the past and confront previously unspoken truths. Given the web's capacity to facilitate representation and celebration of marginalized identities, technical and format considerations guided the selection of an appropriate platform. Unlike downloadable apps, web platforms offer immediate access without requiring downloads, aligning with the project's goal of simplicity and immediacy.

The choice of web streaming format depended on sonic quality and behavior. One way to approach synchronized playback would have been to download samples and use web timing [Borch, 2023] to tightly synchronize across devices with near-zero latency. Ultimately, we decided that prefetching audio would be too slow and that WebRTC would provide "enough" synchronization. The unknown sub-second latency inherent in WebRTC would be a speculative constraint on audio composition, working with the latency's choral-inducing effect among multiple devices instead of against it. This latency or slight WebRTC slippage is conceptually significant for memorializing the 49 human beings, as it introduces a human-like variability rather than a computational precision achieved through prefetching audio.

Our future work will explore haptics and geolocation, aiming to use the Web Vibration API (currently in development) to combine tactile feedback with streaming and enhance the Memorial's accessibility. Participants will ideally hold devices to their chest to feel vibrations. Additionally, we will integrate the Geolocation API to show the relative proximity of connected participants through multiple dots in the interface [Cáceres and Reilly, 2022]. As a memorial experience, drawing participants together in a shared moment of reflection and remembrance, the aim is to continue to grow this

²Broadcast code: https://github.com/augustblack/pulse_memorial

sense of proximity, shared space, and community that the web provides.

The Pulse Memorial project offers a virtual memorial that diverges from traditional, costly monuments known for their density and site-specific focus. While this virtual format enhances accessibility, it raises questions about whether easy access diminishes significance or value. Critics may view the memorial as too inexpensive or lacking permanence. However, the project challenges these perceptions by leveraging digital media to reach broader audiences and foster dynamic, participatory community experiences. It underscores the evolving nature of memorialization in the digital age, inviting exploration of how technology can preserve memory and engage communities meaningfully.

5 Conclusion

This paper introduces Pulse Memorial, a participatory art installation that uses multichannel web audio streaming, data sonification, and field recordings to address queer grief. Drawing inspiration from Toni Morrison's concept of "disremembrance," the project confronts the erasure of queer lives with an immersive sonic experience that encourages collective remembrance and introspection. By highlighting the power of collaborative, participatory, non-hierarchical sonic playback, the project pushes back against contemporary desensitization regarding mass shootings and violence.

In conclusion, we harness mobile technology to facilitate collective remembrance, prioritizing community interaction and participatory listening on-site. The dedicated web broadcast prioritizes networking at a local level so that anyone ingroup scenario can enact the piece without relying on site specificity. It also encourages a kind of "togetherness" where the piece can only be heard when with others. As mass shootings become increasingly frequent, the need for memorials to honor those lost in these tragic events grows. The Pulse Memorial project seeks to find transformative power in digital media as a way to preserve cultural memory and amplify voices that may otherwise be marginalized. It invites continued exploration at the intersection of technology, identity, and remembrance, paving new paths for memorials and community engagement.

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